

Visualization 1- Map of authors

Goal: Show which author's voices from the world are amplified and which are not.

Description: The idea is to scale different countries on the map to make the countries with the most published authors bigger.

Tools:

- GeoJSON/TopoJson: to render a map using D3
- [d3-cartogram](#) : to scale a map given some data about countries
- pandas: to clean up data and create dataset

Lectures:

- D3 (4.2), Data lecture (4.1), Interactions (5.1)

Extra ideas: Allow users to hover and zoom as well as navigate the map. Add a mode to switch from representing nationalities to representing women's percentage

scale countries by amount of books published by author's from there



Visualization 2 - Gender exploration

Goal: Explore gender disparity using different attributes.

Description: The idea is to allow the user to choose different filters and see an interactive chart change given these filters.

Tools:

- D3: to create the interactive chart
- pandas: to clean up data and create dataset

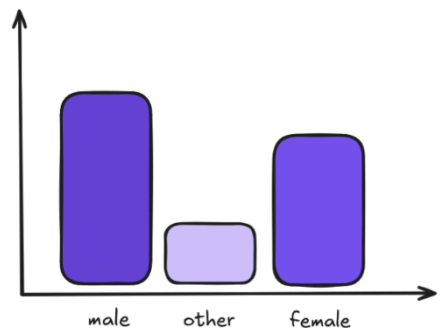
Lectures: D3 (4.2), Data (4.1), Interactive D3 (5.2)

Filters

Countries:

- Asia
 - China
 - India
 - ...
 - Europe
 - Switzerland
 - ...
 - ...
- Genres :
- Romance
 - Fantasy
 - ...

Chart representing gender distribution



Visualization 3 - The library

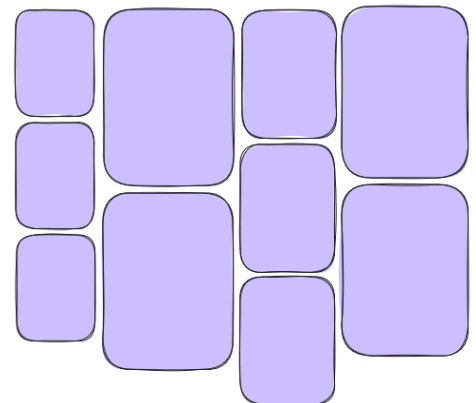
Goal: Allow users to explore the dataset with a non-standard point of view, i.e. non-american and non-male.

Description: The user can filter for a genre of a book, and publishing period and it shows books written by non-male authors not from the US. Hovering above the book gives info about the rating, the votes. We could have one cover color/icon per genre to break uniformity.

Tools: CSS grid for layout, D3 for filtering

Lectures: Data visualisation (1.1), D3 (4.2), Interaction (5.1), Interactive D3 (5.2)

Extra ideas: Allow filtering by publishing period, show the books in a bookshelf



Womens book explorer

Visualization 4 - Price plots

Goal: To uncover correlations between the liked percentage of a book to other factors, specifically analyzing if readers tend to give higher ratings to longer, more expensive books or cheaper ones. Is not limited to price, can be reworked to include other parameters that could reveal more interesting characteristics.

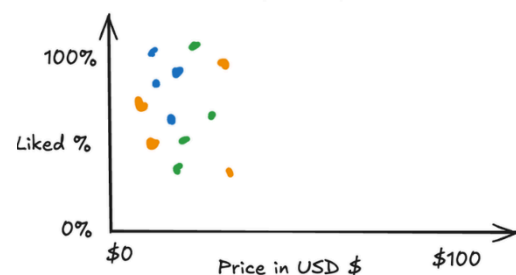
Description: A 2D scatter plot plotting individual books as data points. The x-axis represents the pages field (a Quantitative Ratio variable), and the y-axis Price, but we could also think of having more meaningful data such as number of ratings. We can also link all of the data points together by different colours for a category such as: author, genre, or gender of author, just to name a few.

Tools:

- pandas: to clean up data and create dataset
- D3

Here for example we could have color split by genre or some other data category

Scatter plot of price to liked



Lectures: D3 (4.2), Interaction (5.1), Data (4.1)

Visualization 5 - Bestsellers timeline

Goal: Let users visualize a parallel timeline of bestsellers by gender.

Description: We have a central timeline and users can focus on a year by hovering. When hovering, additional info appears. Also, some more information about that year could appear, like the number of other bestsellers sold by gender.

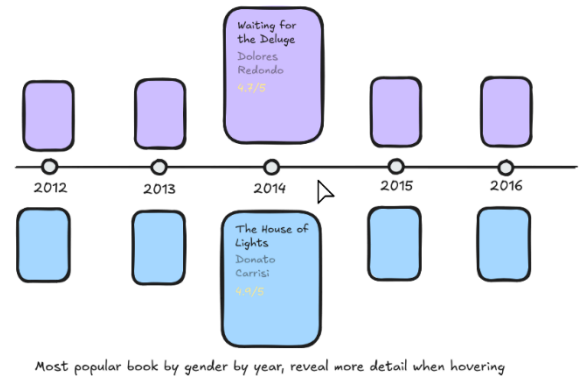
Tools:

D3 for the timeline axis

CSS transitions for expand

pandas for pre-extracting the best sellers per year.

Lectures: Data visualisation (1.1), D3 (4.2), Interaction (5.1), Interactive D3 (5.2).



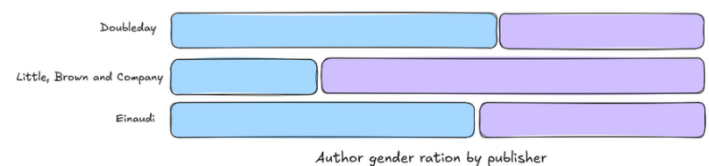
Visualization 6 - Publishers and gender

Goal: Highlight discrepancies between publishers regarding gender

Description: We have one row per publisher, next to it a ratio bar. The rows are sorted by number of publications.

Tools: D3 bar plot

Lectures: Data visualisation (1.1), D3 (4.2), Interaction (5.1), Interactive D3 (5.2).



Visualization 7 - Genres and gender

Goal: Visualize what genres are the most common and correlations to author's gender

Description: We have one circle per genre. The most common the genre is in the dataset, the bigger the circle. The user can select Male/Female and look at the result after a transition.

Tools: D3 circular pack → [Circular Packing | the D3 Graph Gallery](#)

Lectures: D3 (4.2), Interaction (5.1), Interactive D3 (5.2).



Visualization 8 - Published books by country inhabitant

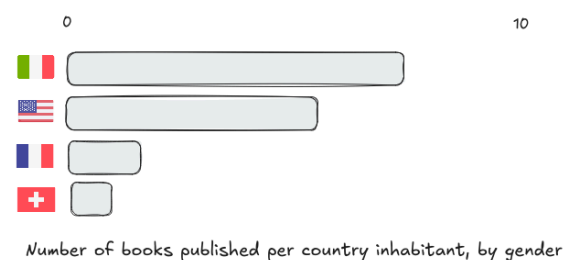
Goal: Put the number of published books in perspective of the size of a country and gender.

Description: One line per country, the lines are sorted by count. The user can select one or more genres to include.

Tools: D3 bar chart.

Lectures: D3 (4.2), Interaction (5.1), Interactive D3 (5.2).

Extra ideas: Let users see how that value evolves over time with a [Bar Chart Race](#).



Visualization 9 - Prizes and gender

Goal: Let the user have an overview of the distribution of literary prizes when conditioning on gender. There could also be a selector on year to see the potential evolution.

Description: Two square shaped clusters representing the amount of prizes.

Tools: D3 circle packs.

Lectures: D3 (4.2), Interaction (5.1), Interactive D3 (5.2).

